

Document 01 — Problem, Use Case & Solution

Station imbalance as a barrier to mode choice

What the problem is, who the tool is for, the data it stands on, and the three concrete things it lets a reviewer do.

Read first. This document frames the decisions that Documents 02–04 explain in depth.

Companion documents: 02 (data & methods), 03 (dashboard), 04 (process & lessons).

Prepared: July 2026 · Independent portfolio project · public data only.

1. The one question

Every part of this project exists to answer a single question:

FRAMING QUESTION

Where and when does station imbalance — no bikes to pick up, or no docks to return to — create a physical barrier to choosing bike share, and how can NYC DOT prioritize rebalancing and expansion investment using data it already collects?

The emphasis on *barrier to choosing* is deliberate. An empty or full station is usually discussed as an operations nuisance — a metric for the rebalancing team to chase. This project reframes it as an **access** problem: a rider who reaches for a bike and finds none, or arrives and finds nowhere to dock, has just been told — physically — that bike share is not reliably available to them. If that happens often enough at the stations serving a given neighborhood, the mode stops being a real option for the people there. That is a transportation-equity question, not only a logistics one.

2. Why imbalance is hard to see, and why that matters

The central measurement difficulty shapes the entire design, so it is worth stating plainly and up front:

THE INVISIBLE-DEMAND PROBLEM

Trip data cannot record the trips that *did not happen* because a station was empty when someone wanted a bike, or full when they wanted to dock. Those are exactly the failures that matter most for access — and they leave no row in the trip archive. Chronic imbalance therefore **understates** the true access problem, rather than overstating it.

Two consequences follow, and both are built into the tool rather than hidden:

- **Net flow is a proxy, not a ground-truth failure count.** The historical layer measures *demand pressure* — the imbalance the demand pattern would create absent perfect rebalancing — not observed empty-minutes. It is an honest, reproducible proxy, and it is labeled as one wherever it appears.
- **A live layer complements the historical one.** The dashboard also reads the public real-time feed (GBFS) so a reviewer can see the actual current fill state of every dock, which is the closest thing to observing the failure directly.

3. Who this is for

The intended reader is the **NYC DOT Bike Share & Shared Mobility team**, and more broadly anyone deciding where rebalancing effort and network-expansion dollars should go. The tool is built to support three real decisions that audience makes:

Rebalancing prioritization

Which stations, in which hours, are chronically short of bikes or short of docks — so overnight pre-positioning and daytime rebalancing can be aimed where the pressure actually is.

Expansion & equity targeting

Whether the stations under the most pressure are the ones serving equity-priority populations (near NYCHA developments and schools, or in transit-gap areas far from a subway entrance).

Intervention sizing

What a change would plausibly do before committing to it — more trucks, a weather scenario, a different equity threshold — tested against the real data, in the browser.

4. The data the project stands on

Everything is derived from **public data, with no API keys and a \$0 hosting stack**. That constraint is a feature: it means the entire analysis is reproducible by anyone and cannot depend on data the DOT team does not already have or cannot obtain.

Source	What it provides	Role in the tool
Citi Bike trip archives (S3)	Every completed trip: start/end station, timestamps, rideable type	The historical net-flow signal — the map's core layer
Citi Bike GBFS (real-time feed)	Live per-station bikes & docks available; station roster & capacity	The live dock-status layer, and station capacity for elasticities
NYC Open Data	NYCHA development footprints; DOE school locations	Per-station equity context (proximity to NYCHA / schools)
NY State Open Data	MTA subway entrances & exits (2024)	Per-station transit-gap context (distance to nearest subway)
Open-Meteo (historical archive)	Real daily temperature & precipitation, by location	Weather features for the model and the weather-scenario elasticities

All sources are public and key-free. Dataset IDs and endpoints are documented in `pipeline/`.

SCALE OF THE COMMITTED DATASET

12 months of real trips — July 2025 through June 2026, a full contiguous year — aggregated across **2,516 stations** in the station roster. The weather regression alone is fit on **788,979** real per-(station, date) observations spanning **435 distinct daily temperatures** from -14.3°C to 32.3°C . These are counts from the committed data files, not estimates.

5. The core signal, in one paragraph

For every station, hour of day, day type (weekday / weekend), and calendar month, the pipeline computes **net flow = arrivals – departures**, normalized to a per-day rate (so a month's many weekdays and few weekend days are comparable). A persistently **negative** morning value means the station *drains* — riders arrive to find no bikes (a deficit / supply failure). A persistently **positive** value means it *fills* — riders arrive to find no open docks (a surplus / dock-scarcity failure). That single signed number, sliced by time and season, is what the map colors. Document 02 explains exactly how it is built and validated.

6. The solution — three things the tool lets you do

The prototype is not a report; it is an interactive instrument. It supports the three decisions from Section 3 through three capabilities.

6.1 See where and when imbalance concentrates

A map of every station, colored by net flow, with a 24-hour time slider, a weekday/weekend toggle, and a season/month selector. Slicing by time is not decoration: NYC bike-share demand in January runs at roughly half of July's amplitude, and warm-month evening peaks run later — so a year-round average would forecast both months wrong. The same map switches to a **live dock-status** layer that reads the real-time GBFS feed at view time.

6.2 Rank and prioritize — with equity context attached

The dashboard surfaces the stations under the most pressure right now as two ranked lists (*most bikes needed*, *most docks needed*), and attaches per-station equity context to every station: its nearest NYCHA development, school, and subway entrance, with real distances. A permanent readout keeps the population-level counts in view — for example, at the default thresholds, how many stations sit within 300 m of a NYCHA development or school, and how many are more than 800 m from any subway entrance.

A DELIBERATE SCOPE BOUNDARY

Equity context is shown **per station** — in the detail panel, in tooltips, in the ranked lists — but the tool intentionally has **no map-level "equity filter"** that recolors or hides the whole map by equity status. This is a standing design decision (recorded in the project's working rules), not an omission. The goal is to let imbalance be *read against* the access agenda, not to reduce the map to a single equity toggle.

6.3 Simulate an intervention before committing to it — "Investigator Mode"

A what-if sandbox lets a reviewer test three kinds of change against the real data, entirely in the browser (every control reads a precomputed file — nothing re-runs a model live):

- **Fleet size** (1–10 rebalancing trucks) → how many deficit/surplus stations get serviced.
- **Weather scenario** (temperature & precipitation, or a preset) → how each station's flow is projected to shift.
- **Equity thresholds** (NYCHA/school proximity, subway gap) → how the flagged population changes, recomputed live.

A running diff bar summarizes baseline-vs-scenario for all three at once, and any scenario can be saved to a shareable URL. Document 03 walks through the interface; Document 02 explains what each control is actually computing.

7. A preview of the honest findings

Document 02 reports the results in full, but three of them define the project's character and are worth stating here, because the tool was built to surface them rather than hide them:

Finding	What it means
The simple baseline wins	A season-aware seasonal-naive forecast beats both a spline GAM and a tuned gradient-boosting model out-of-sample (mean MAE 1.95 vs 2.18 and 2.10 bikes/day, statistically significant). The complex model is not adopted, because it did not earn its complexity.
More trucks help linearly, not with diminishing returns	At this network's scale, every truck hits its own single-shift stop cap long before running out of flagged stations — so the binding constraint is per-truck stop budget, not overall demand. A counter-intuitive result, checked against the data rather than assumed.
Weather effects are real but modest	Warmer days lift flow magnitude and rain suppresses it, at plausible sizes — but the projection is an elasticity estimate from history, explicitly <i>not</i> a weather-specific model, and says so.

8. What this deliberately is *not*

- **Not an official product.** It is an independent, self-directed prototype, not affiliated with or endorsed by NYC DOT or Lyft / Citi Bike.
- **Not a real-time operations console.** The historical analysis is a planning and prioritization instrument; the live layer is a current-state view, not a dispatch system.
- **Not a ground-truth failure meter.** Net flow is a demand-pressure proxy; the tool says so wherever the number appears, and pairs it with the live layer for the real current state.
- **Not a black box.** Every derived number is reproducible from public data by code in `pipeline/`; the models are deliberately auditable (see Document 02).